Evaluating Perfect Square and Cube Roots
Answer Key

Grade 8 / Pre-Algebra

Quick Answers

1. 7
2. 3
3. 12
4. 4
5. 11
6. 15
7. 5
8. 8
9. 13 ft
10. 8 in
11. $>$
12. (a) the corrected value = 4, —

What is verified

11 of 12 problems fully machine-checked against SymPy.
— marks an answer only you can judge — problem 12. The worked solution states what a correct response must contain.

Curriculum

Level: Grade 8 / Pre-Algebra
8.EE.A – *problems 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12*
Difficulty 1–5 of 5 across 13 tagged checks.

Common wrong answers

12. *If they got 8: took the square root of the number instead of the cube root*

1. Evaluate $\sqrt{49}$.
Solution: Look for the number that multiplies by itself to give the radicand: $7 \times 7 = 49$. $\boxed{7}$

2. Evaluate $\sqrt[3]{27}$.
Solution: A cube root asks for the factor used three times: $3 \times 3 \times 3 = 27$. $\boxed{3}$

3. Evaluate $\sqrt{144}$.
Solution: $12 \times 12 = 144$, so the square root is the side of a 12-by-12 square. $\boxed{12}$

4. Evaluate $\sqrt[3]{64}$.
Solution: Test small cubes: $3^3 = 27$ is too small, $4^3 = 4 \times 4 \times 4 = 64$. $\boxed{4}$

5. Evaluate $\sqrt{81} + \sqrt[3]{8}$.
Solution: Evaluate each root first, then add.

$$\begin{aligned} \sqrt{81} &= 9 && \text{since } 9 \times 9 = 81 \\ \sqrt[3]{8} &= 2 && \text{since } 2 \times 2 \times 2 = 8 \\ 9 + 2 &= 11 \end{aligned}$$

The roots are evaluated before the addition. $\boxed{11}$

6. Evaluate $\sqrt{225}$.
Solution: $15 \times 15 = 225$; a useful landmark between $\sqrt{196} = 14$ and $\sqrt{256} = 16$. $\boxed{15}$

7. Evaluate $\sqrt[3]{125}$.
Solution: $5 \times 5 \times 5 = 125$. $\boxed{5}$

8. Evaluate $3\sqrt[3]{216} - \sqrt{100}$.
Solution: The coefficient 3 multiplies the root, it does not go inside it.

$$\begin{aligned} \sqrt[3]{216} &= 6 && \text{since } 6 \times 6 \times 6 = 216 \\ 3 \times 6 &= 18 \\ \sqrt{100} &= 10 \\ 18 - 10 &= 8 \end{aligned}$$

Multiply first, then subtract. $\boxed{8}$

9. A square patio covers exactly 169 square feet of ground. How long is one side of the patio?
Solution: For a square, area = side × side, so the side is the square root of the area: $13 \times 13 = 169$. $\boxed{13 \text{ ft}}$

10. A solid cube-shaped crate holds exactly 512 cubic inches. How long is one edge of the crate?
Solution: For a cube, volume = edge × edge × edge, so the edge is the cube root of the volume: $8 \times 8 \times 8 = 512$. $\boxed{8 \text{ in}}$

11. Write $<$, $>$, or $=$ on the blank to make the statement true.

$$\sqrt{64} \quad \underline{\qquad} \quad \sqrt[3]{216}$$

Solution: Evaluate both roots before comparing: $\sqrt{64} = 8$ because $8 \times 8 = 64$, and $\sqrt[3]{216} = 6$ because $6 \times 6 \times 6 = 216$. Since 8 is to the right of 6 on the number line, the left side is larger. $\boxed{>}$

12. Challenge. Kara evaluates $\sqrt[3]{64}$ and writes 8. (a) Evaluate $\sqrt[3]{64}$ correctly. (b) Explain what Kara did wrong.
Solution (a): A cube root needs the factor used three times: $4 \times 4 \times 4 = 64$. $\boxed{4}$
Solution (b) — instructor-judged. Kara took the *square* root of the radicand instead of the cube root: $8 \times 8 = 64$ is a square-root check, not a cube-root check. A

complete answer names that swap and shows a multiplication test — $4 \times 4 \times 4 = 64$ works, while $8 \times 8 \times 8$ is far larger than the radicand, so 8 cannot be the cube root. Accept any wording that (i) identifies the square-root-for-cube-root confusion and (ii) backs it with a multiplication check.